

Hermenegildo Jerónimo dos Santos

Software Engineer · Full-Stack & AI, Real-Time & Human-in-the-Loop Systems

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SUMMARY

Software engineer building real-time systems across the stack, from mobile to AI, with 5+ years shipping production software across healthcare, enterprise and live events. Primary engineer on Ovee, a live-event AI layer with an operator approving every contribution before it reaches the stage, delivered at 4 live events across Asia and Europe, including a Fortune-500 leadership tour and mci group's CheckedIn 2026, with zero off-message moments on stage. Generalist across backend, frontend, mobile, cloud and AI integration. Stack: TypeScript · React · Python · Go · Azure OpenAI · Node.js.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go, Python, Dart

Frontend / Mobile: React, Next.js, React Native (Expo/Reanimated), Flutter, Three.js, Tailwind CSS, Material UI, Redux

Backend: Node.js, NestJS, FastAPI, Express, GraphQL, WebSockets, MQTT, WebRTC, RESTful APIs

AI / LLM: Azure OpenAI GPT-4o, OpenAI Realtime API, TTS/STT, LLM prompt engineering, human-in-the-loop systems

Data: PostgreSQL, MongoDB, Cosmos DB, Firebase, Redis, MySQL; TypeORM, Prisma, SQLAlchemy

Cloud & DevOps: Azure (Functions, App Service, OpenAI), AWS (EC2, S3, Lambda), Cloudflare Workers, Docker, GitHub Actions

EXPERIENCE

Full Stack Engineer | Dorier

Jun 2025 – Present

Geneva, Switzerland · Remote

Ovee: Live-Event AI (Fortune-500 leadership tour; mci group's CheckedIn 2026)

- Primary engineer (~600 of ~670 commits): operator approval workflow for the moderated stage, FastAPI services, Azure OpenAI GPT-4o generation and TTS voice, React operator and audience apps, WebRTC relay for OpenAI Realtime, WebSocket broadcast
- Delivered at 4 live events across Asia and Europe: a three-summit leadership tour for a Fortune-500 multinational, plus mci group's CheckedIn 2026 in Geneva, where Ovee was billed as co-moderator alongside the group CEO and powered a workshop breakout; 0 off-message moments on stage
- CheckedIn, 3 event days: ~175,000 requests, 7 server errors, 16–42 ms average response; shipped fixes and a second simultaneous workshop room overnight between sessions
- First summit: an audience spike overwhelmed the S1 plan; scaled live to P2v2 and recovered within ~1 minute, then moved to P2v3 and introduced pre-event load testing
- Workshop synthesis: 60 roundtables / 600 leaders turned into themes on the main display in ~5 s; 699 backend test functions

UN Geneva Visitor Centre: Together immersive experience (Fondation Portail des Nations)

- Software engineer on the tour management system, taken over from Local Projects through a nine-week handoff (Jul–Sep 2025) and carried to the public opening in June 2026 (~200,000 visitors/year expected; 3 pavilions, 8 languages, up to 40 visitors per session)
- React Native audio guide, guide tablet and Go tour services over MQTT: in-app Quuppa positioning, audio drift correction (5 s check, 400 ms threshold), headphone reconnection recovery, device-aware tour ownership, group language flows
- Deaf accessibility mode: audio pipeline runs muted so captions stay timed to the show; Gathering proximity effect mapping indoor position (5 m → 0.5 m) to a live visual cue
- Emergency audio end to end (DB migrations, Go service and API, CMS, MQTT, handheld ducking and looping evacuation, offline assets), built and tested, not yet enabled onsite; wrote most automated tests and the platform's technical documentation

Scopely Global Celebration Day: WebAR Character Hunt (sole engineer)

- Only engineer on a browser AR game (no app install) run at 12 Scopely hubs, Tokyo to Mexico, on the same day; designed for ~2,100 players at an event Scopely reported at 2,400 employees
- Built the Mattercraft/Zappar experience, Azure Functions API, Cosmos DB model partitioned by location and the React staff dashboard; five-week core build (Jul 7–Aug 8, 2025) plus three weeks of refinement
- On-device anti-cheat: normalised iOS/Android motion sampling (30/60/100 Hz) with raw-sensor fallback; shake penalised only with walking evidence and ≥200 ms persistence, tuned for fairness

- Server-side score clamp, 12-location whitelist, profanity filter, 5 submissions/IP/location/hour rate limit, 7-day automatic data deletion (GDPR)

Museum AI Concierge: working prototype (voice, vision, grounded answers)

- Primary engineer (~100 of 118 commits): Azure AI Foundry, AI Search, Speech and App Service stack; grounding ladder (catalogue → allow-listed web with citation → known facts → “I don’t have that”) with a one-token LLM relevance gate against keyword over-grounding
- Two model agents (Guide, Monitor) plus deterministic strategy and gamification rules that run only when engagement drops; live probe campaign (57 behavioural + 44 edge cases: prompt injection, malformed input, concurrency, rate limits) and 474 automated tests

Enterprise Automation: Galderma / Lightcraft

- Python CLI/GUI automation tool (Dropbox, Gmail, OAuth2) with cross-platform GitHub Actions builds; in production

Project Development Manager / Full Stack Developer | NomadEngenuity

Apr 2024 – Apr 2025

Covilhã, Portugal · Hybrid

- Project manager for a team of four: client conversations, sprint planning and hiring
- Seezy, a multi-partner eye-care plan platform: co-designed the architecture, then took over the whole system and carried it, front end to back end, to delivery; ~10 NestJS microservices behind Azure API Management on Container Apps, an orchestrator owning a 21-state care-plan process
- Multi-tenant partner access with Auth0 (admin / manager / user per organisation, branches, custom roles, rollback on failed registration); MongoDB/Cosmos DB; email and SMS notifications; CI/CD with GitHub Actions
- Project-managed PharmaSee (staffing platform for pharmacy, optical and audiology professionals, live at pharmasee.pt) and FactorAI, an industrial AI venture for small manufacturers co-pitched at the INNOCUP competition at UBI Medical (Nov 2024); incubated, and its first prospect ROPRE is now a client partner

Software Developer | DC Tech

Mar 2023 – Jun 2023

Covilhã, Portugal · Part-time

- Built award-winning full-stack environmental disaster prevention prototype (React, Node.js, ML models for flood/wildfire prediction); secured incubator funding in Porto

Programming Teacher | Conservatório de Música da Covilhã

Oct 2022 – Jun 2023

Covilhã, Portugal · Part-time

- Designed curriculum (Scratch, HTML/CSS, JavaScript) for 100+ students; built browser-based code editor with real-time preview and gamification

Software Developer | 2GF Innovation Technology

Jul 2021 – Oct 2022

Remote

- Built React/TypeScript interfaces for cross-border family platform, WebRTC video communication, reusable component library, Flutter/Dart offline-first mobile app, Firebase auth/realtime DB
- Developed Node.js middleware integrating SAP enterprise systems; ReactJS dashboards for operational metrics; MongoDB analytics schemas

SELECTED PERSONAL PROJECTS

[InvoFlow](#) · FastAPI · Next.js 14 · PostgreSQL · Azure Document Intelligence · Cloudflare R2 · Clerk · Docker

- Full-stack invoice/receipt management SaaS for Portuguese SMBs: OCR-powered data extraction, real-time IVA tracking, accountant-ready export; Docker Compose deployment with mock mode for local dev

[NexTool API](#) · TypeScript · Cloudflare Workers · Hono · Vitest (30 tests) · RapidAPI

- 13+ developer utility endpoints (QR codes, scraping, hashing, geolocation, JSON tools) on Cloudflare edge; monetised on RapidAPI

EDUCATION

MBA (ESG Focus) · University of Beira Interior Executive, Portugal

Oct 2024 – Jun 2025

BSc Computer Science and Engineering · University of Beira Interior, Portugal · Final project: React + Flutter + Firebase RBAC badge system

Sep 2018 – Jan 2022

LANGUAGES

Portuguese (native) · **English** (full professional proficiency, spoken and written)